

Week 9-10

Autonomy

Building up autonomy

week 8-10

1. Camera integration
2. Autonomous Yaw
3. Autonomous Z (height)
4. Autonomous X,Y
5. FULL Autonomous flight (X,Y,Z,Yaw)
6. FULL Autonomous flight in 15cm sphere

- Milestones 1-6 are last milestones, 30 points total, All due by end of class week 10.
- Please have at least one partner pass the flight milestone before moving to autonomy

Y control

- Goal: Have quad keep Y value of position fixed.
 - Desired Y position should be directly under tag.
 - Note: when $\text{Yaw} \approx 0$, roll controls Y position.
- Solution: Add More PD controllers!!
 - Before: desired roll was based on joystick
 - Now: Desired roll is proportional to distance from desired Y value, and to the Y velocity of the robot.
 - So: Desired roll is now:
$$P_camera(y_camera - \text{Desired_y}) + -D_camera(Y_camera - Y_camera_prev)$$

Y control

- Now there is a
 - PID controller commanding motors to keep robot at desired roll angle (you built this in previous classes).
 - PD controller setting desired roll based on camera Y info
 - Be sure to update D from camera ONLY when a new data point comes in (use sequence number)
 - Hint: before using motors check to see that the desired roll angle makes sense for different camera values
 - Hint: camera sensor has some noise, so you may want to use an exponential filter for x and y values:
 - $\text{camera_y_estimated} = \text{camera_y_estimated} * .6 + \text{camera_y} * .4$
 - You should have a function `camera_control()` to compute desired thrust, pitch, roll, yaw from camera data

Mixed Autonomy

- We want to be able to override the autonomy, or leave it be.
- $\text{Desired_roll} = \text{Desired_roll_from_joystick} * 0.5 + \text{Desired_roll_from_camera} * 0.5$

X control

- Same as y, but with pitch

Milestone 4

- X,Y,yaw control (with mixed autonomy)
- You can choose to use auto thrust or not
- In ground effect, robot should be able to stably hold x,y position.

Milestone 5

1. FULL Autonomous flight (X,Y,Z,Yaw)
 - Should be better than human flight

Milestone 6

1. FULL Autonomous flight (X,Y,Z,Yaw)
 - Stably hold position within 15 cm sphere.

